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Shrikrishna Lolla

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Education

Carnegie Mellon University — Pittsburgh, PA
Master of Science in Machine Learning | Advisor: Prof. Virginia Smith | GPA: 4.0/4.0 — *Dec 2027*

Birla Institute of Technology and Science, Pilani — Hyderabad, India
Bachelor of Engineering in Electronics and Communication | Minor in Data Science | GPA: 4.0/4.0 — *Aug 2023*

Research and Publications

Fully Unsupervised Self-debiasing of Text-to-Image Diffusion Models — WACV 2026
*Korada Sri Vardhana, **Shrikrishna Lolla**, Soma Biswas* | Accepted, Poster

Continual Concept Erasure and Restoration of Diffusion Models — BMVC 2026
Shrikrishna Lolla, Karthikeya Vemuluri*, Korada Sri Vardhana*, Soma Biswas* | Accepted, Poster

PRISM: Precise Concept Suppression for Transformer Based Diffusion Models — AAAI 2027
Shrikrishna Lolla, Korada Sri Vardhana*, Karthikeya Vemuluri, Soma Biswas* | Under Review

Does Quantization Compromise Concept Erasure in Diffusion Models? — WACV 2027
*Korada Sri Vardhana, **Shrikrishna Lolla**, Soma Biswas* | Under Review

Experience

Graduate Student Researcher — Pittsburgh, PA
Prof. Virginia Smith | Carnegie Mellon University — *Aug 2026 – Present*

- Implemented and evaluated scalable influence function based functional attribution, measuring their effectiveness for detecting covert LLM behavior against black-box auditing baselines.

Pre-Doc Fellow @ IACV Lab — Bengaluru, India
Prof. Soma Biswas | Indian Institute of Science — *Nov 2024 – July 2026*

- Proposed SelfDebias for unsupervised, test-time debiasing for Stable Diffusion via hierarchical clustering of h-space latent, **outperforming supervised methods**. Accepted (Poster) @ WACV 2026. Link
- Introduced CORE-Bench to benchmark continual unlearning and restoration, and proposed counter-edit formulations for selectively re-enabling erased concepts. Accepted @ BMVC 2026.
- Developed an inference-time subspace method (PRISM) for concept suppression in FLUX. Analysed where, how, and to what extent to intervene in DiTs and improved erasure runtime by **16x**. Under review @ AAAI 2027.
- Investigated the effect of quantization on unlearning in Diffusion Models across erasure methods, bit-widths and operation orders (pre and post erasure). Under review @ WACV 2027.

Microsoft — Bengaluru, India
Software Engineer | Build Global — *July 2023 – Oct 2025*

- Deployed an ARIMA model that helps save up to **20 hours** during Windows OS build failures.
- Upgraded Perl version from 5.14 to 5.38 on Windows OS to address **200+ CVEs** and improve buildtime by **5%**.
- Won the **best project award** in Azure E+P IDC during the **Microsoft Global Hackathon 2023** by developing a Microsoft Teams feature for the visually impaired using Florence-2 and text-to-speech models.
- Developed a Word add-in using fine-tuned BERT to detect document confidentiality.

JP Morgan Chase — Mumbai, India
Software Developer Intern | MRGR-COO — *Jan 2023 – June 2023*

- Developed and maintained model database services and testing repository.
- Created web dashboards for visualising model testing data and automated report generation for financial services models, enhancing efficiency across business teams.

Microsoft — Hyderabad, India
Software Engineer Intern | OneNote India — *May 2022 – July 2022*

- As part of the OneNote India team, implemented oEmbed functionality for the OneNote Android App by studying its endpoints across Web, iOS, and Desktop versions and proposing an Android-compatible solution.
- Integrated WebView as a part of the native code common across platforms and incorporated key features like

repositioning, resizing, and appropriate zoom and scroll behaviour. Final Presentation

Ziroh Labs Bengaluru, India

Summer Intern *June 2021 – July 2021*

- Designed a Privacy-Preserving Recommender System for Fully Homomorphic Encryption (FHE) recommenders.
- Formulated a quadratic activation after experimenting with existing activation function approximations.
- Developed an improved baseline model using Matrix Factorization. Designed a Neural Graph Collaborative Filtering model to recommend on implicit data. Experimented with GNN recommenders on bipartite graphs.

Projects

FireNet-V2 | *Keras, Python* | GitHub | Report May 2022

- Fire detection using lightweight CNNs on portable devices. Original model: FireNet.
- Utilized Batch Normalization, ELU activation, and increased Dropout to improve accuracy from 93.24% to 96.23% with a marginal increase in size from 7.45MB to 11.7MB.
- Performed data augmentation to compensate for the small trainset size (871 images). Compared against EfficientNetB0, MobileNetV2, ShuffleNetV2 and the original FireNet model.

FCFS and Round Robin Scheduling | *C, Multi-Threading* | GitHub | Report Nov 2021

- Implemented FCFS and Round Robin Scheduling for multiple threads using IPC mechanisms (semaphores, pipes, shared memory).
- Analysed performance by plotting charts for turnaround time and waiting time for different processes and scheduling algorithms.

Multipath Video Streaming | *Network Programming, Python* | GitHub | Report April 2021

- Developed a video streaming application using MPQUIC and MPTCP protocols, streaming across networks and on loopback at a consistent frame rate of 20–30 fps.
- Tested the application against packet loss, bandwidth throttling, and drop in connection.

Image Clustering with Autoencoders | *Keras, Python* | GitHub Oct 2020

- Studied the effectiveness of U-Net, Convolutional, and Variational Autoencoders for image clustering: extracted latent space representations to reduce dimensionality and clustered them using K-means.

Skills

Languages and Frameworks: C++, C, Python, Java, JavaScript, CSS, C#, PyTorch, Tensorflow, Keras | **Coursework (BITS):** Object Oriented Programming, Operating Systems, Cryptography, Foundation of Data Science, Information Retrieval, Machine Learning, Data Mining, Network Programming | **Coursework (CMU):** 10-715* (Advanced Topics in Machine Learning), 36-700* (Probability and Mathematical Statistics)

Achievements

ACM ICPC Amritapuri 2020: Preliminary Round: Rank 504/4393, Regional Round: Rank 430/722 teams, Honorable Mention | **Merit Scholarship** 7/8 semesters (top 1% of batch) | **INSPIRE Fellowship** by Dept. of Science & Technology, Govt. of India (top 1% in 12th Board)

Volunteer Activities

CRUx – The Programming and Computing Club of BITS Hyderabad Hyderabad, India

- **Machine Learning Summer Group (MLSG):** Introduced ML to over 100 participants through a beginner-friendly curriculum, assignments, and personalized feedback.
- **Data Structures and Algorithms Workshops:** Curated content, practice problems, and contests to help final year students in their job interviews.
- **Problem Setter:** Created problems and generated test cases for BITS Hyderabad's competitive programming event, CodeDash 2022.

BPHC Ultimate Hyderabad, India

- **South Central Sectionals – 2022:** Hosted the South Central Sectionals in BITS Hyderabad. Took care of the logistics and hospitality of external teams arriving at our campus.
- **Hat tournaments – 2019:** Hosted hat tournament for 60+ players from Hyderabad. Helped in planning the tournament, creating diverse and balanced teams.